Coherence Sweet Spots Are Frequency Minima in Disguise: Metric Sensitivity in Transmon and Fluxonium Design Spaces

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Abstract—That dielectric loss scales with transition frequency is well known, and it implies that a lower-frequency qubit relaxes more slowly at fixed quality factor. We show that this has a sharper consequence for design studies than is usually acknowledged: under a common noise model, modelled T_2 over the transmon (E_J, E_C) plane correlates with f_{01} at $r = -0.9999$ within the transmon regime, and over the fluxonium (E_J, E_L) plane at -0.9973. Each plane carries one effective degree of freedom, so a raw T_2 maximum locates the lowest-frequency corner of the swept domain rather than a design, and both of our own headline optima prove to be grid-edge points outside any usable frequency band. Controlling for frequency changes the conclusions. The transmon has no interior optimum in E_J/E_C: at fixed f_{01}, T_2 is flat to 4.8% across a tenfold range while the anharmonicity falls 7.3×, so charge insensitivity is bought with gate speed rather than coherence, and any reported optimum sits on whichever regime threshold is imposed. The fluxonium does have a genuine interior optimum, visible only once second-order $1/f$ flux dephasing is included, but it lies at 10–75 MHz where $\bar{n}_{\mathrm{th}} = 4$–$32$. For the cross-architecture question we find no single answer: the ratio spans ∼700× across defensible metrics, and at matched frequency the T_2 advantage inverts to a median 0.86× across the literature noise band. A lab-frame pulse simulation settles the widest of those metrics at 36–58×, excluding both analytic bounds, because the fluxonium's drive couples 10.2× more strongly to its leakage transition than to its qubit transition.

Index Terms—superconducting qubits, transmon, fluxonium, decoherence, $1/f$ noise, sweet spots

I. Introduction

Superconducting qubit design is routinely discussed in terms of sweet spots: bias points at which the transition frequency is stationary with respect to a noise coordinate, so that first-order dephasing vanishes. The transmon achieves this by exponential suppression of charge dispersion [1]; the fluxonium achieves it at half a flux quantum, where $\partial\omega_{01}/\partial\Phi = 0$ by symmetry [2], [3]. Both are then said to have a coherence sweet spot in design space as well, and the point of best modelled T_2 over a swept parameter plane is reported as an optimum.

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Code, data and figures: https://github.com/ishitabera/quantum-coherence-sweet-spots-transmon-fluxonium

This paper argues that the second usage does not survive scrutiny, and that the error is systematic rather than incidental. We address three questions:

- Q1 Where are the coherence optima across the transmon (E_J, E_C) and fluxonium (E_J, E_C, E_L, Φ) design spaces?
- Q2 Which noise channel dominates where?
- Q3 Does the fluxonium's flux sweet spot buy more robustness than the transmon's charge insensitivity, and at what cost?

Our central methodological claim is that Q1 and Q3 cannot be answered without controlling for the transition frequency, and that once controlled the conventional answers change sign or dissolve. Our central negative result is that the fluxonium's coherence advantage at matched frequency does not survive the uncertainty in the noise amplitudes that the model itself assumes.

Figure 1 summarises the study. All results are model-based estimates under literature noise amplitudes. They are not predictions of device performance, and Section IX quantifies how far they sit from measurement.

II. Related Work and Contribution

A. What is already established

The frequency scaling we build on is not new. Dielectric loss from two-level-system defects at surfaces and interfaces is the dominant relaxation channel in transmons, with the relaxation rate approximately proportional to the surface participation ratio [4]; at fixed quality factor the rate grows with ω_{01}, so a lower-frequency qubit relaxes more slowly. Design-space optimisation under noise models is likewise established: Papič et al. identify optimal transmon parameter ranges through comprehensive noise modelling including charge-parity effects [5], and Ram et al. give a staged framework spanning circuit design, spectral analysis and calibration for both architectures [6]. Experimentally, fluxonium coherence has advanced well past the devices we calibrate against: Somoroff et al. report $T_2 = 1.43$ ms with average gate fidelity 0.99991 [7].

Two works are close enough to this one to be worth locating precisely.

McRae et al. [8] argue that coherence metrics cannot currently be compared across measurements, devices and

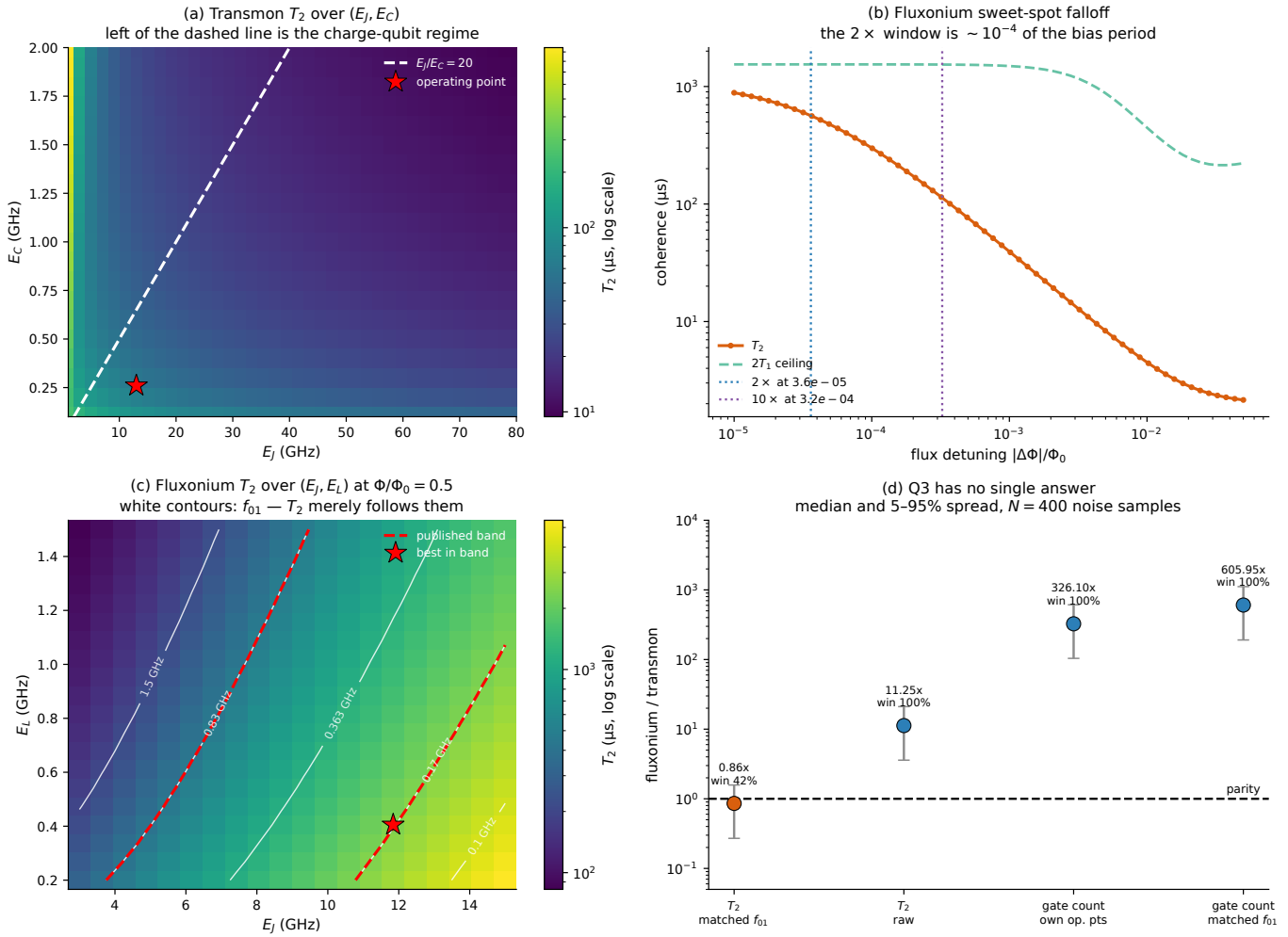


Fig. 1. The study in four panels. (a) Transmon T_2 over (E_J, E_C) at $n_g = 0$; left of the dashed line is the charge-qubit regime, whose numbers are not transmon results. (b) The fluxonium sweet-spot falloff on a log-detuning axis, with the $2T_1$ ceiling and the two- and tenfold loss points; on a linear flux axis the entire collapse hides inside the first 0.1% of the plot. (c) Fluxonium T_2 over (E_J, E_L) at half flux, with f_{01} contours in white — the T_2 structure merely follows them, which is the confound this paper is about. (d) The Q3 answer under each metric, with the 5–95% spread over $N = 400$ noise-amplitude draws; orange marks the metrics in which the fluxonium wins fewer than 90% of samples.

laboratories, because temporal fluctuation, TLS dynamics and unreported experimental detail dominate the differences being claimed. Their concern is experimental reproducibility. The present paper makes the modelling analogue of that argument: a comparison drawn from a swept design space is equally unreliable unless the control variable and the figure of merit are stated with the number.

Lange et al. [9] compare transmon and fluxonium lattices directly, and— importantly—do so under an explicit control, designing each architecture for similar gate speed and fidelity before comparing. Their figures of merit are residual ZZ cross-talk and state localisation; coherence is not among them. Our work is complementary: same instinct about controlled comparison, different quantity.

B. Contribution

Our contribution is not the frequency scaling but its consequence for how design studies are read:

1) We quantify the confound as a near-complete rank deficiency of the design plane in both architectures ($|r| > 0.99$), so that a raw T_2 optimum is a statement about where the sweep stopped rather than about a design.
2) We show the transmon optimum is pinned to whatever regime threshold is imposed, across eight thresholds spanning $E_J/E_C = 15$–60, so the threshold and not the optimisation is the substantive choice.
3) We show that including second-order $1/f$ flux dephasing creates an interior fluxonium optimum that a first-order model—the standard tooling default—does not see at all.
4) We show the architecture comparison spans two orders of magnitude across defensible metrics, and settle the widest of those metrics by direct pulse simulation.

We are not aware of a study reporting the metric sen-

TABLE I
Noise channels by architecture.

Channel	scqubits method	Tmon	Flux
Dielectric loss (T_1)	t1_capacitive	✓	✓
Quasiparticle (T_1)	t1_quasipart...	–	✓
$1/f$ flux, 1st order	tphi_1_over_f_flux	–	✓
$1/f$ flux, 2nd order	this work	–	✓
Charge / offset	tphi_1_over_f_ng	✓	–
Critical current	tphi_1_over_f_cc	✓	✓

sitivity of the transmon–fluxonium coherence comparison explicitly. That statement rests on the survey above, which is not exhaustive; the individual ingredients are all known, and the claim is about their combination.

C. A caveat on our own control

Matching f_{01} removes the dominant confound but not all of it. Dielectric loss also depends on geometry and surface participation, which differ between architectures and which a Hamiltonian-level model does not capture [4], and real devices fluctuate in ways no static model represents [8]. Frequency matching is necessary, not sufficient, and our matched-frequency numbers should be read as removing one large confound rather than as a fully controlled experiment.

III. Model and Conventions

A. Noise channels

Coherence is estimated with scqubits [10]. Rates add,

$$\frac{1}{T_2} = \frac{1}{2T_1} + \sum_k \frac{1}{T_{\phi,k}}, \tag{1}$$

and channels are selected per architecture. A plain transmon has no flux loop, so $1/f$ flux dephasing is not a supported channel for it; the fluxonium has no charge dispersion to speak of. Table I lists the assignment.

Noise amplitudes are $Q_{\mathrm{cap}} = 10^6$, $A_{n_g} = 10^{-4}$, $A_\Phi = 10^{-6}\,\Phi_0$, $A_{cc} = 10^{-7}$; the flux amplitude sits at the optimistic end of the measured range, which clusters near 1–$2\,\mu\Phi_0$ across flux-sensitive devices [11], [3]. Temperature, quasiparticle density and the $1/f$ cutoffs are the scqubits defaults ($T = 15\,\mathrm{mK}$, $x_{qp} = 3 \times 10^{-6}$, $t_{\mathrm{exp}} = 10\,\mu\mathrm{s}$) and are recorded with every result, since they set the answers and are easily left undeclared.

B. Second-order flux dephasing

scqubits implements $1/f$ dephasing to first order in $\partial\omega_{01}/\partial\lambda$, which vanishes identically at a sweet spot; the package returns a divergent time there, documented behaviour rather than a bug [10]. The physical rate at half flux is second order, and we implement the closed form used for fluxonium by Nguyen et al. [3], after Ithier et al. [12]:

$$\Gamma_\Phi^{(2)} = \frac{\partial^2\omega_{01}}{\partial\Phi^2}\, A_\Phi^2. \tag{2}$$

Equation (2) carries no logarithmic factor. Other treatments attach a prefactor of order $|\ln(\omega_{\mathrm{low}} t_{\mathrm{exp}})|$ or its square, which moves the answer by two orders of magnitude; we report the convention dependence rather than burying it. The required curvature is Richardson-extrapolated, because a plain central difference is still 4% from converged in the deep-well regime where this channel matters most.

C. A declared modelling choice

Passing Q_{cap} explicitly installs a flat, frequency-independent quality factor in place of the scqubits default $Q_{\mathrm{cap}}(\omega) \propto \omega^{-0.7}$ [13]. At the fluxonium sweet spot the two differ by 7.1× in T_1, making this the largest single modelling choice in the study. The spread is one-sided rather than symmetric: Nguyen et al. fit their own T_1-versus-frequency data with $\tan\delta_C \propto \omega^{0.15}$, and at $f_{01} = 0.363\,\mathrm{GHz}$ the flat model sits 1.52× from their measured exponent while the scqubits default sits 4.7× from it. We retain the flat value as both the conservative and the better-supported choice, and declare it.

IV. Validation

Forty-four checks establish the pipeline before any result is generated. They compare against exact Hamiltonian symmetries (machine precision), basis-size convergence, analytic limits from Koch et al., and—in seven cases—data measured on real devices.

The sharpest analytic anchor is Koch's worked charge-dephasing example: at $E_J = 30\,\mathrm{GHz}$, $E_C = 0.35\,\mathrm{GHz}$ and $A = 10^{-4}$ he quotes $T_2 \sim 8\,\mathrm{s}$; feeding our computed charge dispersion through his Eq. (5.9) gives 8.75 s. The maximum charge-dispersion derivative reproduces his $1.5 \times 10^{-4} E_C$ at $E_J/E_C = 50$ to a ratio of 0.830, the expected sign for an asymptotic expression approached from below.

Against Manucharyan et al. [2], the lowest two transitions at $\Phi_{\mathrm{ext}} = 0$ are reproduced to better than 0.7% under every one of that paper's three published parameter sets, and a single four-parameter fit reproduces all six digitized lines across two flux biases to an RMS of 0.0102 GHz. We note that those three parameter sets—the rounded main-text values and two Table I least-squares fits—span 9.5% in f_{01} at the flux sweet spot, so agreement quoted against the main-text values alone overstates the constraint.

The third transition at $\Phi_{\mathrm{ext}} = 0$ is excluded from this claim for a physical reason rather than a loose tolerance. Its bare value lies within 0.14 GHz to 0.36 GHz of the junction-array mode that Manucharyan's extended model places at 10.78 GHz, with a 0–3 coupling near 0.53 GHz: the coupling exceeds the detuning by 1.5–3.7×, so the line is strongly hybridized and no bare-circuit model can reproduce it. Explicit diagonalization of the fluxonium coupled to both array modes confirms that this is a $\Phi_{\mathrm{ext}} = 0$ effect and bounds what our ideal-superinductance model gives up by omitting them: the dressed f_{01} shifts by −16% at $\Phi_{\mathrm{ext}} = 0$ but by only

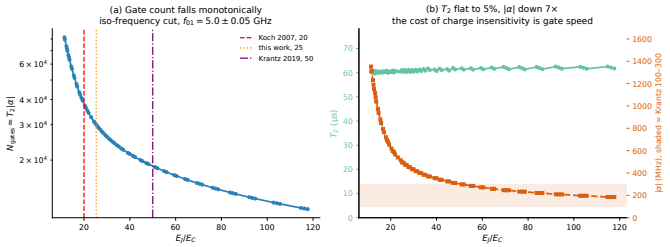


Fig. 2. The transmon trade-off at fixed frequency ($f_{01} = 5.00 \pm 0.05$ GHz). (a) Gate count falls monotonically with E_J/E_C; the three vertical lines are candidate regime thresholds, and the optimum sits on whichever one is imposed. (b) T_2 is flat to 4.8% while $|\alpha|$ falls 7.3×; the shaded band is the Krantz 100 MHz to 300 MHz design window. Charge insensitivity costs gate speed, not coherence.

−0.43% at the flux sweet spot, where every coherence number in this study is evaluated—smaller than the 9.5% spread across the paper's own published parameter sets.

Finally, the second-order channel of Eq. (2) is the one piece of physics implemented here rather than taken from established tooling, so it is validated against a published number rather than against its own formula. Evaluated on Nguyen's nine devices at their measured $A_\Phi = 1.8\,\mu\Phi_0$ it yields $T_\phi^{(2)} = 15$–804 ms, clearing their stated bound of > 10 ms for all qubits on every device.

V. The Transmon Design Plane

We sweep $E_J \in [1, 80]$, $E_C \in [0.1, 2.0]$ GHz at $n_g = 0$ and $n_g = 0.25$ (800 points each), and refine to a 65×53 grid over $E_J \in [4, 20]$, $E_C \in [0.08, 0.6]$ GHz.

A. The confound

Within the transmon regime $E_J/E_C \geq 20$, $\mathrm{corr}(\log T_2, \log f_{01}) = -0.9999$. The plane contains one degree of freedom, not two. The reason is structural rather than numerical: dielectric loss dominates T_1 everywhere in the regime, with $\Gamma \propto |\langle 0|\hat{n}|1\rangle|^2 S(\omega_{01})$. Since $|\langle 0|\hat{n}|1\rangle|^2 \propto \sqrt{E_J/E_C}$ and $S \propto E_C$, the rate goes as $\sqrt{E_J E_C} \propto \omega_{01}$—a function of frequency alone. Consequently the best T_2 point of the swept plane, 187 μs at $E_J = 3.03$, $E_C = 0.10$ GHz, sits on two grid edges at $f_{01} = 1.45$ GHz, below the 3 GHz to 6 GHz band real transmons occupy [14]. It is a frequency minimum, not a design.

B. No interior optimum

Holding $f_{01} = 5.00 \pm 0.05$ GHz and sweeping E_J/E_C from 11.4 to 117.6, T_2 is flat to 4.8% (59.7 μs to 62.6 μs) while $|\alpha|$ falls by a factor of 7.3 (1353 MHz to 185 MHz). Charge insensitivity is bought with anharmonicity—that is, with gate speed—and essentially not with coherence at all (Fig. 2).

That the trade is anharmonicity-for-charge-insensitivity is expected from Koch's asymptotics: with $\alpha \simeq -E_C$ and $f_{01} \simeq \sqrt{8E_J E_C}$ held fixed, raising E_J/E_C forces $E_C \propto (E_J/E_C)^{-1/2}$, predicting a factor 3.21 over our swept range against a measured E_C ratio of 3.47. The observed 7.3× fall in $|\alpha|$ is steeper still because $\alpha \simeq -E_C$ is asymptotic and fails at the low end: $|\alpha|/E_C = 2.29$ at $E_J/E_C = 11.4$ against 1.09 at 117.6. The trade-off is therefore harshest exactly in the low-E_J/E_C corner that an unconstrained gate-count metric wants to select, which is an independent reason to keep the design envelope.

Relocating the gate-count optimum under eight regime thresholds spanning $E_J/E_C = 15$ to 60, the optimum tracks every one of them to within 0.24. It sits on the constraint, never inside the domain. This is the signature of a constrained optimum, and it means the substantive choice in this part of the design space is the threshold, not the optimisation. The correct claim is a monotone trade-off terminated wherever the regime boundary is drawn, and any quoted optimum is meaningful only together with its threshold. We adopt the Krantz envelope ($E_J/E_C \geq 50$, $|\alpha| \in [100, 300]$ MHz, $f_{01} \in [3, 6]$ GHz) [14] because it is independent of every noise amplitude in the study, giving the operating point $E_J = 13.0$ GHz, $E_C = 0.26$ GHz, $f_{01} = 4.92$ GHz, $T_2 = 62.4$ μs, $|\alpha| = 299$ MHz.

A limitation of the model bears on this choice: tphi_1_over_f_ng covers $1/f$ dephasing only, and the broadband charge sensitivity that motivates Krantz's $E_J/E_C \geq 50$ is absent. Left to itself the metric under-penalizes low E_J/E_C and would recommend $|\alpha| \approx 728$ MHz, roughly 2.4× what is fabricated. The envelope is a constraint imposed on the model, not a result derived from it.

VI. The Fluxonium Design Plane

Over $E_J \in [3, 15]$, $E_L \in [0.2, 1.5]$ GHz at $E_C = 2.5$ GHz and $\Phi/\Phi_0 = 1/2$, the same confound appears: $\mathrm{corr}(\log T_2, \log f_{01}) = -0.9973$. The unconstrained maximum, 5427 μs, sits at both grid edges at $f_{01} = 75$ MHz, where $\bar{n}_{\text{th}} = 3.7$—thermally scrambled and 2.3× below the lowest published device. Constrained to the published band 0.17 GHz to 0.83 GHz the best design gives 2412 μs at $E_J = 11.8$, $E_L = 0.405$ GHz.

A. An interior optimum, created by second-order dephasing

Extending the scan to $E_J = 45$ GHz reveals a genuine interior maximum in T_2, present at every (E_C, E_L) combination tested (Fig. 3). It is created by the second-order flux channel: as E_J rises and E_L falls the qubit sinks into a deep double well, f_{01} collapses, and $\partial^2 f_{01}/\partial\Phi^2$ grows by three orders of magnitude across the scan, so $\Gamma_\Phi^{(2)}$ grows with it and eventually overwhelms the improving dielectric T_1. At the turning point that channel carries 43–58% of the total dephasing rate, which is the signature of a genuine internal optimum: the channel that turns the curve over is comparable to everything else exactly at the turn. A first-order-only pipeline shows no turnover over the same range and rises monotonically to 2.3×10^5 μs—which is precisely how an earlier version of this analysis concluded there was no interior optimum, and is a concrete argument

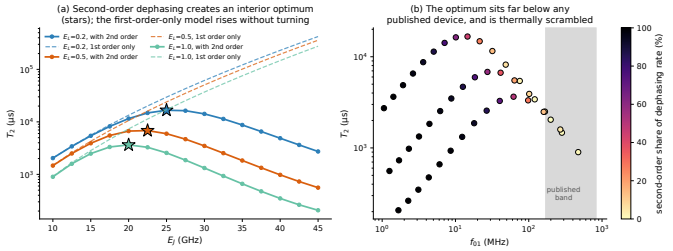


Fig. 3. (a) T_2 versus E_J at three values of E_L. Solid curves include second-order flux dephasing and turn over (stars); the dashed first-order-only curves rise without limit over the same range, which is how the interior optimum was previously missed. (b) The same points against f_{01}, coloured by the second-order share of the dephasing rate. Every turning point lies far below the published fluxonium band (shaded).

for including the second-order channel in any design study that sweeps toward low E_L.

The mechanism is geometric. As E_J rises and E_L falls the potential minima separate, the tunnel splitting that sets f_{01} collapses, and the same deepening well sharpens the curvature $\partial^2 f_{01}/\partial\Phi^2$, which grows by three orders of magnitude across the scan. Since $\Gamma_\Phi^{(2)} \propto \partial^2\omega_{01}/\partial\Phi^2$ by Eq. (2), second-order dephasing grows with it and eventually overwhelms the improving dielectric T_1. The transmon has no analogue: its charge dispersion falls exponentially in $\sqrt{E_J/E_C}$ with no turning point.

The optimum is nevertheless inaccessible. It occurs at $f_{01} = 10$–$75\,\mathrm{MHz}$ with $\bar{n}_{\mathrm{th}} = 4$–$32$. Past the turnover the frequency correlation reverses sign, which is a further indication that the second-order channel introduces genuinely new structure rather than rescaling the old.

VII. Dominant Noise Channels

At a charge-sensitive bias ($n_g = 0.25$) charge noise dominates the transmon below $E_J/E_C \approx 25$ and dielectric relaxation above it. That crossover is measured from our own data and lands close to Koch's independently derived bound of 20; the two arguments are unrelated—one a noise-budget crossover, the other a pulse-duration requirement—so their agreement is genuine corroboration. Critical-current noise never dominates anywhere in 1600 points at literature amplitudes. The crossover is noise-model dependent (it spans 20.8–29.9 across literature ranges) and is therefore reported as a result rather than used as an input.

For the fluxonium, relaxation dominates at the sweet spot and first-order flux noise becomes the largest single term beyond $2.75 \times 10^{-5}\,\Phi_0$. Quasiparticle tunnelling is not a dead channel: it is suppressed only at the sweet spot, where f_{01} is minimal, and limits T_1 at most detuned points. This finding is conditional on $x_{qp} = 3 \times 10^{-6}$, a default that Nguyen et al. measure to be 300× lower for their devices; at their density the channel is negligible everywhere. It is the most fragile result in the study and is reported as conditional.

VIII. The Cross-Architecture Comparison

A. Bias robustness, the one unambiguous asymmetry

Both bias coordinates are periodic with period one, so the width of the window over which T_2 stays within a factor k of its sweet-spot value, as a fraction of one bias period, is directly comparable. The transmon's T_2 varies by 1.0056× across its entire charge period—a factor-of-two loss never occurs at any bias. The fluxonium loses a factor of two within $\pm 3.61 \times 10^{-5}\,\Phi_0$ and a factor of ten within $3.25 \times 10^{-4}\,\Phi_0$, located by root-finding on the model rather than read off a grid. The usable bias window differs by 1.4×10^4.

The falloff exponent is not a single number. It approaches the first-order $1/f$ prediction of -1 only in an intermediate window where T_2 is genuinely flux-dephasing limited, reaching -0.956 there, and flattens on both sides—capped by the $2T_1$ ceiling close in, and contaminated by other channels far out.

Intrinsic $1/f$ flux noise does not by itself drive the qubit off the spot: even at Nguyen's measured amplitude integrated over an hour, the RMS wander leaves a 24× margin to the tenfold-loss detuning. The binding constraint is systematic drift and bias-line calibration, not stochastic noise—which relocates the difficulty rather than removing it, and makes it scale with qubit count.

This constraint is being engineered away, which we should say plainly. Bénâtre et al. hold gradiometric fluxonium qubits at their sweet spot passively, reporting stability of $3 \times 10^{-4}\,\Phi_0$ from target with only a 15% reduction in echo coherence [15]. Their achieved stability sits essentially at our computed tenfold-loss detuning of $3.25 \times 10^{-4}\,\Phi_0$, yet they observe far less degradation than that number would suggest—unsurprisingly, since their device has different E_L and hence different curvature, and echo suppresses low-frequency drift that free induction does not. We read their result as evidence that the bias-precision cost we quantify is real, understood, and addressable by design rather than by calibration alone. It weakens the scalability pessimism in our conclusion without touching the arithmetic behind it.

B. No single ratio

Table II gives the comparison under every metric we consider defensible, and Fig. 4 shows how each fares across the noise band.

Two ranges are visible, and they are of comparable size. Across metrics the answer spans roughly 700×. Within the gate-count metric alone it spans 393×, depending on what bounds the minimum pulse length. The $|\alpha|$ bound—leakage to $|2\rangle$ only—is the textbook choice, but for a fluxonium at half flux $|\alpha| = 8.7\,\mathrm{GHz}$ is 24× the transition frequency it would drive, so that bound licenses a pulse 24× shorter than one period of the transition itself. The drive-period bound $1/f_{01}$ is equally physical and gives 1.34×.

TABLE II
Fluxonium/transmon ratio by metric. MC = median over $N = 400$ log-uniform draws of each noise amplitude across its literature decade, with the 5–95% sample spread.

Metric	Det.	MC med. [5–95%]	Wins
Raw T_2	18.1	11.3 [3.6, 21.1]	100%
T_2, matched f_{01}	1.1–2.2	**0.86** [0.27, 1.6]	42%
Gates, $1/\lvert\alpha\rvert$	524	326 [104, 611]	100%
Gates, $1/\min(\lvert\alpha\rvert, f_{01})$	22.0	13.7 [4.4, 25.6]	100%
Gates, $1/f_{01}$	**1.34**	**0.83** [0.26, 1.6]	40%
Gates, simulated	48	–	–
Bias window, 2× loss	1.4×10^4 wider for the transmon		

TABLE III
Shortest calibrated X gate reaching an average error below 10^{-3}, from the lab-frame simulation, and the resulting gate count $N = T_2/T_{\text{gate}}$. Durations are upper bounds and are bracketed by the duration grid and by optimiser convergence.

	T_{gate} (ns)	T_2 (µs)	N
Transmon	8.0 (6.0–8.0)	62.4	7,803
Fluxonium	3.0 (2.5–3.0)	1129.7	376,571
Simulated ratio			**48**×
bracket from grid + optimiser			36–58×
for comparison, $1/\lvert\alpha\rvert$ bound			524×
for comparison, $1/f_{01}$ bound			1.34×

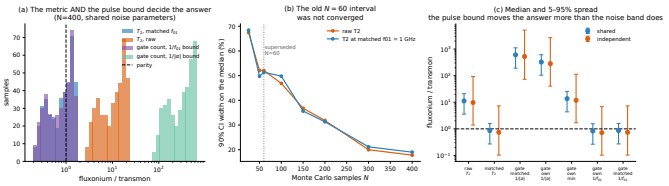


Fig. 4. Robustness of the Q3 answer to the noise amplitudes. (a) Distribution of the fluxonium/transmon ratio under four metrics, $N = 400$ shared-parameter draws; the metric and the pulse bound each move the answer by more than the noise band does. (b) The 90% bootstrap CI on the median against sample size, showing that an earlier $N = 60$ interval had not converged. (c) Median and 5–95% spread per metric under shared versus independent sampling.

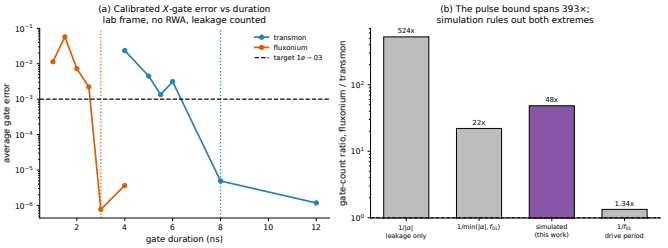


Fig. 5. (a) Average X-gate error against gate duration for both architectures, from the lab-frame simulation with amplitude, DRAG and Stark detuning calibrated at each point; dotted lines mark the shortest duration meeting the 10^{-3} target. Non-monotonicity at the shortest durations is optimiser noise and is not smoothed. (b) The gate-count ratio under each pulse-length bound. The two analytic bounds differ by 393×; the simulation (purple) falls between them and excludes both.

C. The raw advantage is mostly frequency

The two devices differ in frequency by 13.6×, and $\log 13.55/\log 18.10 = 0.90$: about 90% of the log-advantage is the frequency difference. This is not a coincidence of these two designs. Dielectric loss dominates both architectures and its rate scales with ω_{01}, so $T_2 \propto 1/f_{01}$ in both—measured as $f_{01}^{-0.965}$ for the transmon. Because the flux sweet spot is a frequency minimum, the fluxonium is structurally advantaged on any uncontrolled T_2 comparison, and reporting one credits the architecture with what the frequency did.

Controlling for it, the deterministic ratio at matched f_{01} is 1.07–2.16×. But propagated across the literature noise band the median is 0.86× and the fluxonium wins in only 42% of samples. At matched frequency the transmon is more often the better device. This is the most important negative result here: what remains of the coherence advantage after the frequency control does not survive the noise-amplitude uncertainty.

D. Settling the pulse bound

The gate-count range is wide enough to make the metric almost uninformative, so we settle it by simulation rather than by argument. We integrate the lab-frame Schrödinger equation for each architecture with no rotating-wave approximation, using a six-level truncation of the scqubits eigenbasis and a charge drive, and calibrate each gate as an experimentalist would: amplitude, DRAG coefficient [16] and Stark-shifted drive frequency are optimized at every duration, and the average gate error is evaluated on the computational subspace so that leakage is counted; Fig. 5 shows the resulting error curves.

This exposes an effect that the anharmonicity argument misses entirely. The fluxonium's $|\alpha|/f_{01} = 24$ looks decisive, but its drive matrix elements are unfavourable: at half flux $|\langle 1|\hat{n}|2\rangle|/|\langle 0|\hat{n}|1\rangle| = 10.2$, against 1.37 for the transmon. The drive couples ten times more strongly to the leakage transition than to the qubit transition, so the effective figure of merit is the anharmonicity divided by that ratio, not the anharmonicity itself.

Table III reports the shortest calibrated gate reaching an average error below 10^{-3} for each architecture, and the resulting gate count. These durations are upper bounds: a three-parameter calibration over a fixed Gaussian shape is far from optimal control, and GRAPE or CRAB over a free pulse shape would shorten both—not necessarily equally. The point is not the precise value but that the simulated ratio falls decisively between the two naive bounds, excluding both.

The result is that the fluxonium does convert coherence into operations more efficiently than the transmon, but by a factor of order 36–58× rather than the ∼500× the anharmonicity bound suggests or the ∼1× the drive-period bound suggests. Neither extreme survives contact with a simulated pulse, and the residual uncertainty is dominated by optimiser convergence rather than by physics.

IX. Calibration Against Measurement

Run at the parameters of each of the nine devices in the published Table I of Nguyen et al., the model is optimistic by a median factor of 2.0× on T_1 and 2.6× on T_2, worst case 5.7×. The gap is not uniform, so it is not a single missing constant—and it is smallest where it matters most: device I, the device nearest our nominal design in frequency (0.395 versus 0.363 GHz), is matched to 4% on T_2, with the model pessimistic on its T_1.

Identified contributors, in order: the flat $Q_{\mathrm{cap}} = 10^6$ against their measured $\tan\delta_C = (2.0\text{–}3.6) \times 10^{-6}$ at 6 GHz; the absence of Purcell decay, photon shot noise and inductive loss; the fact that their T_2 is a Hahn echo while this model computes free-induction decay, which makes the comparison conservative in the model's favour; and a temperature mismatch, since we run at the 15 mK default while they state 20 mK.

The comparison also has a floor set by our own model rather than by the devices. Absolute T_2 spans 149 µs to 3316 µs under single-decade moves of the noise amplitudes, so a factor of 2.6 is well inside the band the model itself admits; the calibration should be read as establishing the right order of magnitude, not as a residual to be explained away. Consistent with that, the state of the art has moved well past the devices we calibrate against—Somoroff et al. report $T_2 = 1.43$ ms [7], so a model tuned to reproduce Nguyen's 2019 numbers exactly would now be miscalibrated against 2023 hardware.

We stress that the published table must be used. The arXiv v1 preprint lists eight devices and omits device I entirely—the paper's headline device, and the one closest to the design regime studied here.

X. Limitations

The model omits Purcell decay, photon shot noise, inductive loss in the junction array, TLS spectral structure, crosstalk and any lattice; qubits are treated in isolation. The superinductance is ideal, so array-mode hybridization is absent, though Section IV bounds that omission at −0.43% in f_{01} at the flux sweet spot. Second-order dephasing uses a convention with no logarithmic factor; a $\ln^2$ convention would change sweet-spot T_2 by ∼37% instead of 0.6% and would move, though not remove, the interior optimum. Absolute T_2 spans 149 µs to 3316 µs under single-decade moves of the noise amplitudes and should never be quoted without a band. The Monte Carlo varies amplitudes only; fabrication spread in E_J, E_C, E_L is not propagated, and the log-uniform priors are a choice. The gate-count metric is a heuristic, not a literature formula.

XI. Conclusion

Neither architecture has a coherence sweet spot in design space. Both design planes are, to better than $|r| = 0.99$, restatements of the transition frequency, so a raw T_2 maximum locates the edge of the swept domain or the point where a frequency constraint was drawn. The transmon is a monotone trade-off in which charge insensitivity is paid for in anharmonicity, once, at design time. The fluxonium has a real interior optimum, visible only once second-order flux dephasing is included, but it is thermally out of reach; in practice it too is a trade-off terminated by a frequency floor.

For Q3, the honest output is a metric family, not a number. The fluxonium's raw 18× coherence advantage is roughly 90% frequency; at matched frequency it inverts to a median 0.86× across the literature noise band. Its durable advantage is anharmonicity per unit coherence, and its price is operating-point precision—a bias window four orders of magnitude narrower than the transmon's, paid continuously at run time rather than once at design time. Whether that trade is worth making is a question about qubit count and calibration budget, not about coherence.

More generally, we suggest that architecture comparisons in this field should state the metric, the control variable and the uncertainty band together. A single ratio between two architectures at their own operating points is not a comparison between architectures.

XII. Outlook

Two extensions would sharpen the results most. The gate-count metric is currently settled by a coarse calibration; optimal control over a free pulse shape (GRAPE or CRAB) would tighten the 36–58× bracket and, more interestingly, would test whether the fluxonium's unfavourable leakage matrix element can be shaped around—the anharmonicity is genuinely there, and a pulse that exploits it without driving $|1\rangle \to |2\rangle$ would move the number back toward the leakage-only bound. Second, the interior fluxonium optimum we find is thermally inaccessible at 15 mK, but that is a statement about temperature as much as about design; the same analysis at lower base temperature, or with active initialisation modelled, would establish whether it is inaccessible in principle or merely inconvenient.

A broader point follows from the metric sensitivity. If a single architecture comparison can span two orders of magnitude depending on choices that are all defensible, then comparisons reported as single ratios carry less information than they appear to. We would encourage the practice McRae et al. [8] advocate for experimental coherence numbers—report the conditions with the number—to be extended to modelled design comparisons: state the control variable, state the figure of merit, and give the spread.

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with IEEE policy on the use of artificial intelligence in authorship.

Reproducibility and Data Availability

All code, generated data and figures are openly available at https://github.com/ishitabera/quantum-coherence-sweet-spots-transmon-fluxonium.

Every number in this paper is produced by five notebooks run in order from a single validated pipeline; no notebook reimplements the physics. Each sweep writes its raw data before any figure code executes, together with a metadata record of the grid, the noise amplitudes, the software versions and the git revision. Every quoted value is bound to the file and filter that produced it by an automated provenance ledger of 32 entries, which recomputes and asserts each one on every run and fails rather than warns if a value drifts from its artifact. A test suite of 32 checks pins the conventions and the validated anchors, including the published numbers used for calibration.

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